

Focal shunt versus matched injection

focal shunt

descendants attenuated

$$V_i = q_i(E_E - V_i)$$

I_{soma} → z
← δ_0

soma V restored

matched injection control

no conductance change

$$\kappa(E_I - V_k)$$

I_{soma} → z
← δ_0

soma V restored

baseline focal current matched

One exact gain per ancestry block

root 864691135409937097

$q' = \text{diag}(h) B \eta$

$h = \text{baseline transfer}$

descendants (16)

g_{shunt}

δ_0

50 μm

sister blocks (3)

soma side

B

η

Γ_u

0.958

0.978

0.988

0.994

0.998

q'/q

rows collapsed: 17/5/4/6/46 per band

$\kappa = 0.390$; identity residual 1×10^{-17}

driving forces may vary within a block

Figure 3 is a line graph titled "Permissive normalized model". The y-axis is labeled "localization index (log units)" and ranges from 0 to 0.25. The x-axis is labeled "dose ($\times$ local conductance)" and has four discrete points: 0.25, 0.5, 1, and 2. There are two data series: "focal shunt" (green line with circles) and "matched injection" (grey line with squares). Both series show an increase in the localization index as the dose increases. The "focal shunt" series starts at approximately 0.03 at dose 0.25 and rises to approximately 0.19 at dose 2. The "matched injection" series starts at approximately 0.01 at dose 0.25 and rises to approximately 0.07 at dose 2. Error bars representing standard error are shown for each data point.

Dose ($\times$ local conductance)	focal shunt (log units)	matched injection (log units)
0.25	~0.03	~0.01
0.5	~0.06	~0.02
1	~0.10	~0.04
2	~0.19	~0.07

Figure 3 is a line graph showing the localization index (log units) for four substituted factors: inject, d' , q' , and $q'd'$. The y-axis ranges from 0 to 0.25. The x-axis labels are inject, d' , q' , and $q'd'$. Data points are shown as open circles with error bars. Lines connect the data points for each factor across the different conditions. A bracket above the q' and $q'd'$ points indicates a difference of +0.079.

Physical calibration

shunt - injection (log units)

membrane resistance ($\Omega \text{ cm}^2$)

Δ Disjoint, 45 cells
axial/leak 2.7–133

$\blacklozenge$ Initial, 8 cells
axial/leak 1.4–136

near-zero zoom

no contrast

0

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